

RAKSHA-NETRA

EDGE-AI × MULTIMODAL SENSING × CYBERSECURITY

Rapid presumptive trace screening for railway security environments

COLLECT → ANALYZE → ALERT → SECURELY RECORD → REPORT

NATIONAL-LEVEL HACKATHON PROJECT • TECHNICAL PROJECT BRIEF

KLN COLLEGE OF ENGINEERING

01 PORTABLE Rugged handheld field concept	02 EDGE AI Local multimodal inference	03 OFFLINE-FIRST Operate through connectivity gaps	04 SECURE LOG Cryptographically verifiable history
--	--	---	---

01 AT A GLANCE

RAKSHA-NETRA is a proposed rugged, handheld, Edge-AI-enabled presumptive trace-screening platform. It combines VOC, electrochemical and optical sensing with local sensor fusion, on-device inference, secure event recording and dashboard-oriented reporting.

Design principle: keep the critical screening workflow close to the operator while preserving a secure and traceable digital record.

SAMPLE Controlled collection	SENSORS VOC • EC • Optical	STM32 Acquisition & control	RPI 5 Fusion & preprocessing	EDGE AI TFLite inference	SECURE LOG Hash-chain + metadata	DASHBOARD Report & monitor
---	---	--	---	---------------------------------------	---	---

02 WHY THIS ARCHITECTURE

- STM32 handles deterministic, real-time acquisition, heating/control, safety and diagnostics.
- Raspberry Pi 5 provides higher-level preprocessing, multimodal fusion, Edge-AI inference, storage and communications.
- The sensing layer is modular, allowing future sensor cartridges and dataset/model expansion.
- Offline-first storage prevents temporary network loss from stopping the core workflow.

03 MULTIMODAL SENSING STACK

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'SENSING CHANNEL' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8, greek=0, italic=0, link=[], rise=0, text='SENSING CHANNEL', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CORE COMPONENTS' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8, greek=0, italic=0, link=[], rise=0, text='CORE COMPONENTS', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'ROLE' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8, greek=0, italic=0, link=[], rise=0, text='ROLE', textColor=Color(1,1,1,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph
VOC response	BME688 + SGP40	Capture volatile-response patterns
Electrochemical	SPCE + AD5940	Measure electrochemical responses
Optical / colorimetric	TCS34725 + Camera	Capture optical/colorimetric changes

MULTI-MODAL CHEMICAL-RESPONSE FINGERPRINTING

04 EDGE-AI DECISION PIPELINE

1. Acquire synchronized readings from multiple sensing modalities.

- 2. Condition and preprocess individual sensor streams.
- 3. Extract features or embeddings and fuse the modalities.
- 4. Run TensorFlow Lite inference locally on Raspberry Pi 5.
- 5. Present a presumptive classification/alert state to the operator.
- 6. Record the event and supporting metadata through the secure logging layer.

05 SECURITY BY DESIGN

CONTROL	IMPLEMENTATION	VALUE
Confidentiality	AES-256	Protect stored/transmitted sensitive data
Integrity	SHA-256	Detect content modification
Record linkage	Hash-chained events	Make unauthorized sequence changes detectable
Provenance	Device ID + timestamp + GPS	Contextualize each screening event
Availability	Offline-first local storage	Continue operation through connectivity gaps

06 OPERATIONAL FLOW

COLLECT	ANALYZE	FUSE	CLASSIFY	ALERT	RECORD / REPORT
---------	---------	------	----------	-------	-----------------

07 OFFLINE-FIRST + DASHBOARD

When connectivity is unavailable, screening events can remain on-device in protected local storage. When connectivity returns, queued information can be synchronized according to the defined backend policy.

DEVICE → SECURE LOCAL EVENT LOG → CONNECTIVITY → SYNCHRONIZATION → RPF DASHBOARD

08 PROTOTYPE EXPERIENCE

The Figma prototype demonstrates the intended operator journey: device readiness → controlled sample handling → multimodal analysis → presumptive threat state → secure event log → RPF dashboard.

FIGMA PROTOTYPE INSERT FIGMA URL	GITHUB REPOSITORY INSERT GITHUB URL
-------------------------------------	--

09 INNOVATION SNAPSHOT

- Multimodal chemical-response fingerprinting instead of dependence on a single sensing channel.
- Edge-AI inference reduces dependence on continuous cloud connectivity.
- Hardware/software separation supports deterministic control and flexible AI evolution.
- Cryptographically verifiable event history strengthens traceability of recorded screening events.
- Modular architecture creates a pathway for future sensor cartridges and model upgrades.
- Integrated device-to-dashboard workflow turns a sensing prototype into an operational concept.

10 TECHNOLOGY MATRIX

LAYER	TECHNOLOGY
MCU	STM32
Edge computer	Raspberry Pi 5
AI inference	TensorFlow Lite
Sensing	BME688 • SGP40 • SPCE • AD5940 • TCS34725 • Camera
Connectivity	Wi-Fi • 4G/LTE • MQTT / REST

Security	AES-256 • SHA-256 • Hash-chained logs
Design / Prototype	EasyEDA • Figma

11 VALIDATION ROADMAP

PHASE	FOCUS	EVIDENCE
01 • Prototype	Integrate sensing, MCU, Edge computer and UI	Working proof-of-concept
02 • Dataset	Controlled multimodal response collection	Labeled dataset
03 • Model	Training, validation, quantization	Edge-compatible model
04 • Security	Threat modeling and integrity testing	Hardened event pipeline
05 • Field validation	Environmental and operational testing	Performance evidence

12 REPOSITORY MAP

Recommended GitHub structure keeps the README concise while making evidence and engineering assets easy to inspect.

PATH	PURPOSE
Documentation/	Project brief, report, architecture and specifications
Hardware/	Schematic, PCB and BOM
Software/	STM32 and Raspberry Pi implementation
AI_Model/	Model, preprocessing and inference assets
Dashboard/	Dashboard/backend prototype
Prototype/	Figma exports and screenshots
Presentation/	Hackathon presentation

13 RESPONSIBLE USE

RAKSHA-NETRA is a prototype for presumptive screening and decision support. Sensor responses can be affected by environmental conditions, contamination, calibration drift and sample variability. A presumptive result must not be treated as definitive identification; appropriate confirmatory testing, authorized procedures, calibration and independent validation are required before operational deployment.

RAKSHA-NETRA

A secure, portable and intelligent approach to presumptive field screening.

EDGE AI • MULTIMODAL SENSING • OFFLINE-FIRST • CRYPTOGRAPHIC TRACEABILITY